Further Calculus

Questions 1–314

2.23 — Question 1. Express $\dfrac{2}{x(x^2+2)}$ in the form $\dfrac{A}{x}+\dfrac{Bx+C}{x^2+2}$.

2.23 — Question 2. Express $\dfrac{3x^2-x+10}{(x+1)(x^2+1)}$ in the form $\dfrac{A}{x+1}+\dfrac{Bx+C}{x^2+1}$.

2.23 — Question 3. A student is trying to express a rational function in partial fractions. Their working is shown below:

$$\frac{3x^2+5x+5}{(x+1)(x^2+2)} = \frac{A}{x+1} + \frac{B}{x^2+2}.$$

a) Explain the mistake made by the student.

b) Express $\dfrac{3x^2+5x+5}{(x+1)(x^2+2)}$ correctly as partial fractions.

2.23 — Question 4. Write each of the following as partial fractions:

a) $\dfrac{x+5}{(x^2+1)(x+1)}$

b) $\dfrac{5x}{(2x+1)(x^2+1)}$

c) $\dfrac{x-2}{(x+3)(x^2+4)}$

d) $\dfrac{3x^2-6x+15}{(2x^2+1)(x-1)}$

e) $\dfrac{25x^2}{2x^3-3x^2+8x-12}$

f) $\dfrac{x^2}{x^3-3x^2+5x-15}$

g) $\dfrac{1}{(x^2+k)(x-1)}$, where k is a positive constant.

h) $\dfrac{k}{(x^2+k)(x-k)}$, where k is a positive constant.

2.23 — Question 5. It is known that

$$\frac{6x^2+5x+6}{x^3+kx} \equiv \frac{A}{x} + \frac{Bx+C}{x^2+k},$$

where k, A, B and C are positive integers. Determine the possible values of k. **The following questions require the use of Partial Fractions and A-Level Maths Integration Methods.**

2.23 — Question 6. Find each of the following in simplified form:

a) $\displaystyle\int \frac{18}{(x-1)(x+2)(x-4)}\,dx$

b) $\displaystyle\int \frac{x-2}{(x-1)^2}\,dx$

c) $\displaystyle\int \frac{7x}{(2x-1)(x+3)^2}\,dx$

2.23 — Question 7. Show that

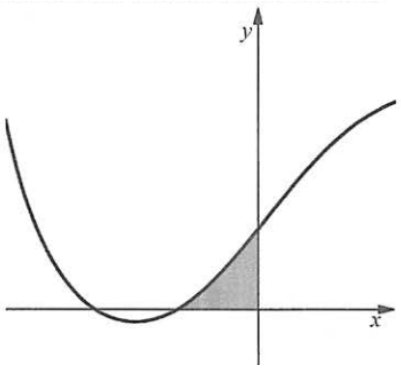
$$\int \frac{3x^2 + x + 4}{(x+1)(x^2+2)}\,dx = \ln\left((x+1)^2\sqrt{x^2+2}\right) + c.$$

2.23 — Question 8. Show that

$$\int_1^4 \frac{x^2 - 4x + 15}{(x+1)(x^2+3)}\,dx = \ln\frac{3125}{722}.$$

2.23 — Question 9. The diagram below shows part of the curve with equation

$$y = \frac{4x^2 + 6x + 2}{(x+2)(x^2+2)}, \qquad x \neq -2.$$



Find the exact value of the shaded area.

2.23 — Question 10. Find the value of k such that

$$\int_4^6 \frac{(k-1)x^2 - 3kx - 4}{(x-3)(x^2+4)}\,dx = \ln\left(\frac{32}{3}\right).$$

2.24 — Question 1. Show that

a) $\dfrac{d}{dx}(\arcsin x) = \dfrac{1}{\sqrt{1-x^2}}$

b) $\dfrac{d}{dx}(\arccos x) = -\dfrac{1}{\sqrt{1-x^2}}$

c) $\dfrac{d}{dx}(\arctan x) = \dfrac{1}{1+x^2}$

2.24 — Question 2. Find the gradient of the tangent to the curve:

a) $y = \arcsin x$ at $x = \frac{1}{3}$

b) $y = \arccos x$ at $x = -\frac{1}{4}$

c) $y = \arctan x$ at $x = \frac{2}{5}$

The following questions use the derivatives of $\arcsin kx$, $\arccos kx$ and $\arctan kx$ and require no further Chain, Product or Quotient Rule.

2.24 — Question 3. Differentiate each of the following:

a) $y = \arcsin 2x$

b) $y = \arccos 3x$

c) $y = \arctan 8x$

d) $y = \arcsin \dfrac{x}{3}$

e) $y = \arccos \dfrac{x}{5}$

f) $y = \arctan\left(-\dfrac{2}{3}x\right)$

2.24 — Question 4. Show that

a) $\dfrac{d}{dx}\left(\arcsin \dfrac{x}{2}\right) = \dfrac{1}{\sqrt{4-x^2}}$

b) $\dfrac{d}{dx}\left(\arccos \dfrac{3x}{4}\right) = -\dfrac{3}{\sqrt{16-9x^2}}$

c) $\dfrac{d}{dx}\left(\arctan \dfrac{5x}{3}\right) = \dfrac{15}{9+25x^2}$

2.24 — Question 5. Find the exact gradient of the curve $y = 8\arcsin\dfrac{x}{4}$ at the point with x-coordinate 2.

2.24 — Question 6. Find the equation of the tangent to the curve $y = 3\arccos x$ when $x = \frac{1}{2}$, writing your answer in the form $y = ax + b$ where a and b are given in exact form.

2.24 — Question 7. Find the equation of the tangent to the curve $y = 2\arccos\frac{x}{3} - \arcsin\frac{x}{3}$ when $x = \frac{3}{\sqrt{2}}$, writing your answer in the form $y = ax + b$ where a and b are given in exact form.

2.24 — Question 8. Find the equation of the tangent to the curve $y = 4\arctan\left(\frac{2x}{5}\right)$ when $x = \frac{5}{2}$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

2.24 — Question 9. Find the equation of the normal to the curve $y = 4 - 3\arctan\dfrac{x}{4} - 2\arctan\dfrac{x}{3}$ when $x = 0$, writing your answer in the form $px + qy = r$ where p, q and r are integers.

2.24 — Question 10. Given $f(x) = \arctan x$ and $g(x) = \arctan 3x$, solve the equation $f'(x) = g'(x)$.

2.24 — Question 11. A curve has the equation $y = \arcsin\left(\frac{2}{3}x\right)$, where $-\frac{3}{2} \leq x \leq \frac{3}{2}$.

a) Show that the gradient of the curve is never equal to $\frac{1}{2}$.

The point P lies on the curve with x-coordinate $-\frac{3\sqrt{2}}{4}$. The normal to the curve at P passes through the y-axis at the point Q.

b) Show that the exact area of the triangle OPQ, where O is the origin, can be written as $\frac{3\sqrt{2}}{64}(9 + 2\pi)$ units^2.

2.24 — Question 12. Show that the curve with equation $y = \arcsin x - \arccos x$ is increasing for $-1 < x < 1$.

2.24 — Question 13. It is given that $y = 3\arctan 6x$.

a) Show that there are no points on the curve where the gradient is equal to 24.

b) State the maximum gradient of the curve.

2.24 — Question 14. Find the exact coordinates of the single stationary point of the curve $y = \arctan 2x - x^2$.

Prerequisite knowledge for Q15: Binomial Series

2.24 — Question 15. It is given that $f(x) = \arctan 3x$.

a) Find $f'(x)$.

b) Assuming that x is small, use your answer to part (a) to express $f'(x)$ in ascending powers of x as far as the term in x^4.

c) Hence express $\arctan 3x$ in ascending powers of x as far as the term in x^5.

2.24 — Question 16. It is given that $f(x) = p\arcsin x - qx$, where p and q are positive constants. Show that the curve $y = f(x)$ has precisely two stationary points when $p < q$.

2.24 — Question 17. Find the values of x for which the curve with equation $y = 5x + \sqrt{21}\arccos x$ is increasing. Write your answer in set notation.

2.24 — Question 18. Show that the curve with equation $y = 3\arctan\dfrac{x}{6} - 2\arctan\dfrac{x}{4}$ has a stationary point of inflection at the origin. **The following questions use the derivatives of** $\arcsin kx$**,** $\arccos kx$ **and** $\arctan kx$ **and use any differentiation knowledge from A-Level Maths.**

2.24 — Question 19. Differentiate each of the following:

a) $y = \arcsin x^2$

b) $y = \arccos \sqrt{x}$

c) $y = \arctan \dfrac{1}{x}$

d) $y = \cos(\arcsin 2x)$ for $-\dfrac{1}{2} < x < \dfrac{1}{2}$

e) $y = \sin\left(\arccos \dfrac{x}{2}\right)$ for $-2 < x < 2$

f) $y = \cos(\arctan 4x)$ for $x \in \mathbb{R}$

2.24 — Question 20. Find the equation of the tangent to the curve $y = (\arcsin x)^2$ when $x = \dfrac{\sqrt{3}}{2}$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

2.24 — Question 21. Find the equation of the tangent to the curve $y = \arctan(e^x)$ when $x = -\frac{1}{2}\ln 3$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

2.24 — Question 22. Differentiate each of the following:

a) $y = x \arcsin x$

b) $y = x^2 \arccos 2x$

c) $y = x^3 \arctan(x^2)$

2.24 — Question 23. It is given that $y = (x-5)\sqrt{-x^2+10x-24} + \sin^{-1}(x-5)$, where $4 \leq x \leq 6$. Show that $\dfrac{dy}{dx} = k\sqrt{-x^2+10x-24}$, where k is an integer to be found.

2.24 — Question 24. It is given that

$$y = 3x\sqrt{4-x^2} + 12\tan^{-1}\left(\frac{x}{\sqrt{4-x^2}}\right), \qquad -2 < x < 2.$$

Show that $\dfrac{dy}{dx} = k\sqrt{4-x^2}$, where k is an integer to be found.

2.24 — Question 25. It is given that a curve has equation $\arccos y - \arccos 3x = \frac{\pi}{3}$, where $-\frac{1}{6} \leq x \leq \frac{1}{3}$. Find the value of $\frac{dy}{dx}$ when $x = \frac{1}{6}$.

2.24 — Question 26. Show that

a) $\frac{d}{dx}(\text{cosec}^{-1} x) = -\frac{1}{\sqrt{x^4 - x^2}}$

b) $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{\sqrt{x^4 - x^2}}$

c) $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1 + x^2}$

2.25 — Question 1. Use the substitution $x = \sin u$ to show that

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + c.$$

2.25 — Question 2. Use the substitution $x = \cos u$ to show that

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = -\arccos x + c.$$

2.25 — Question 3. Find the exact value of

$$\int_{-1/2}^{1/2} \frac{1}{\sqrt{1-x^2}}\,dx.$$

2.25 — Question 4. Use the substitution $x = \tan u$ to show that

$$\int \frac{1}{1+x^2}\,dx = \arctan x + c.$$

2.25 — Question 5. Find the exact value of

$$\int_{-1}^{1} \frac{1}{1+x^2}\,dx.$$

2.25 — Question 6. Find:

a) $\displaystyle\int \frac{2}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{5}{1+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{1-9x^2}}\,dx$

d) $\displaystyle\int \frac{1}{1+4x^2}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{1-2x^2}}\,dx$

f) $\displaystyle\int \frac{1}{1+8x^2}\,dx$

g) $\displaystyle\int \frac{10}{\sqrt{1-5x^2}}\,dx$

h) $\displaystyle\int \frac{200}{1+10x^2}\,dx$

i) $\displaystyle\int \frac{4}{\sqrt{1-\frac{1}{4}x^2}}\,dx$

j) $\displaystyle\int \frac{18}{1+\frac{2}{3}x^2}\,dx$

2.25 — Question 7. Find the exact value of:

a) $\displaystyle\int_0^{\sqrt{2}/8} \frac{1}{\sqrt{1-16x^2}}\,dx$

b) $\displaystyle\int_{1/(2\sqrt{3})}^{1/2} \frac{30}{\sqrt{1-3x^2}}\,dx$

2.25 — Question 8. Find the exact value of:

a) $\displaystyle\int_0^{\sqrt{3}/6} \frac{1}{1+36x^2}\,dx$

b) $\displaystyle\int_{1/\sqrt{3}}^{1} \frac{12}{1+3x^2}\,dx$

2.25 — Question 9.

a) Show that

$$\frac{1}{2+x^2} \equiv \frac{1}{2(1+(kx)^2)},$$

where k is a real number to be found.

b) Hence find $\displaystyle\int \frac{1}{2+x^2}\,dx$.

2.25 — Question 10.

a) Show that

$$\frac{1}{\sqrt{2-x^2}} \equiv \frac{1}{\sqrt{2}\sqrt{1-(kx)^2}},$$

where k is a real number to be found.

b) Hence find $\displaystyle\int \frac{1}{\sqrt{2-x^2}}\,dx$.

2.25 — Question 11. Find:

a) $\displaystyle\int \frac{1}{4+x^2}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{9-x^2}}\,dx$

c) $\displaystyle\int \frac{1}{5+x^2}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{8-x^2}}\,dx$

e) $\displaystyle\int \frac{4}{12+x^2}\,dx$

f) $\displaystyle\int \frac{10}{\sqrt{5-x^2}}\,dx$

g) $\displaystyle\int \frac{1}{2+4x^2}\,dx$

h) $\displaystyle\int \frac{1}{\sqrt{16-9x^2}}\,dx$

i) $\displaystyle\int \frac{2}{4+25x^2}\,dx$

j) $\displaystyle\int \frac{3}{\sqrt{12-\frac{1}{3}x^2}}\,dx$

2.25 — Question 12. Find the exact values of:

a) $\displaystyle\int_{-\sqrt{3}/3}^{\sqrt{3}/3} \frac{1}{\sqrt{4-9x^2}}\,dx$

b) $\displaystyle\int_{0}^{\frac{5}{8}(\sqrt{5}-1)} \frac{3}{\sqrt{100-16x^2}}\,dx$

2.25 — Question 13. Find the exact values of:

a) $\displaystyle\int_{-5\sqrt{3}/27}^{5\sqrt{3}/27} \frac{1}{25+81x^2}\,dx$

b) $\displaystyle\int_{0}^{3(\sqrt{2}-\sqrt{3/2})} \frac{9}{9+2x^2}\,dx$

2.25 — Question 14. Given that a and b are constants, find:

a) $\displaystyle\int \frac{1}{\sqrt{a^2-x^2}}\,dx$

b) $\displaystyle\int \frac{1}{a^2+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{a^2-b^2x^2}}\,dx$

d) $\displaystyle\int \frac{1}{a^2+b^2x^2}\,dx$

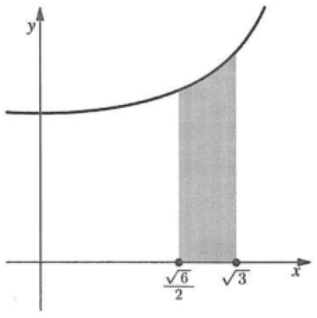
2.25 — Question 15. Given that a and b are positive constants, find the exact value of

$$\int_0^{\sqrt{ab}} \frac{1}{\sqrt{b - \frac{1}{a}x^2}}\, dx.$$

2.25 — Question 16. The curve shown in the diagram below has equation

$$y = \frac{6}{\sqrt{18 - 3x^2}}.$$

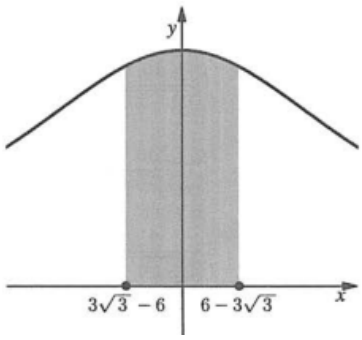
Find the exact area of the shaded region.



2.25 — Question 17. The curve shown in the diagram below has equation

$$y = \frac{9}{36 + 4x^2}.$$

Find the exact area of the shaded region.



2.25 — Question 18. It is given that

$$f(x) = \frac{19x^2 + 34}{(1 + x^2)(9 + 4x^2)}, \qquad x \in \mathbb{R}.$$

a) Write $f(x)$ in the form

$$\frac{Ax + B}{1 + x^2} + \frac{Cx + D}{9 + 4x^2},$$

where A, B, C and D are constants to be found.

b) Hence find $\displaystyle\int f(x)\,dx$.

2.25 — Question 19. Find:

a) $\displaystyle\int \frac{107x^2 - 3}{(4 + 9x^2)(1 + 25x^2)}\,dx$

b) $\displaystyle\int \frac{8x^4 + 12x^3 + 33x^2 + 3x + 4}{(4 + x^2)(1 + 4x^2)}\,dx$

2.25 — Question 20. Show that

$$\int_{1/2}^{\sqrt{3}/2} \frac{16x^2 + 5x + 4}{x(8x^2 + 2)}\,dx = a\pi + \ln b,$$

where a is a rational number and b is a positive integer to be found.

2.25 — Question 21. Given that

$$\frac{1+x}{1+x^2} \equiv \frac{1}{1+x^2} + \frac{x}{1+x^2},$$

find $\displaystyle\int \frac{1+x}{1+x^2}\,dx$.

2.25 — Question 22. Find:

a) $\displaystyle\int \frac{2x+1}{1+4x^2}\,dx$

b) $\displaystyle\int \frac{3+5x}{4+x^2}\,dx$

c) $\displaystyle\int \frac{2-3x}{9+4x^2}\,dx$

d) $\displaystyle\int \frac{5-4x}{25+16x^2}\,dx$

2.25 — Question 23. Given

$$\frac{1+x}{\sqrt{1-x^2}} \equiv \frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}},$$

find $\displaystyle\int \frac{1+x}{\sqrt{1-x^2}}\,dx$.

2.25 — Question 24. Find:

a) $\displaystyle\int \frac{3x-1}{\sqrt{1-9x^2}}\,dx$

b) $\displaystyle\int \frac{1+8x}{\sqrt{4-x^2}}\,dx$

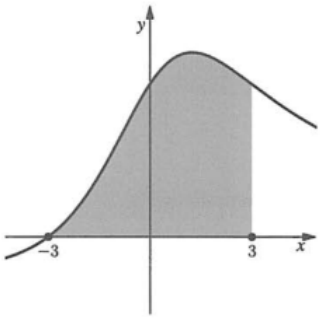
c) $\displaystyle\int \frac{6-x}{\sqrt{9-4x^2}}\,dx$

d) $\displaystyle\int \frac{10-9x}{\sqrt{4-25x^2}}\,dx$

2.25 — Question 25. The curve shown in the diagram below has equation

$$y = \frac{3+x}{9+x^2}.$$

Find the exact area of the shaded region.



2.25 — Question 26.

a) Find the general solution to the differential equation

$$\frac{dy}{dx} = \frac{x+1}{\sqrt{9-x^2}}.$$

b) Hence, find a particular solution such that $y = 3$ when $x = 0$.

2.25 — Question 27.

a) Find $\displaystyle\int \frac{x}{\sqrt{25-x^2}}\,dx$.

b) Find $\displaystyle\int \frac{1}{\sqrt{25-x^2}}\,dx$.

c) It is given that

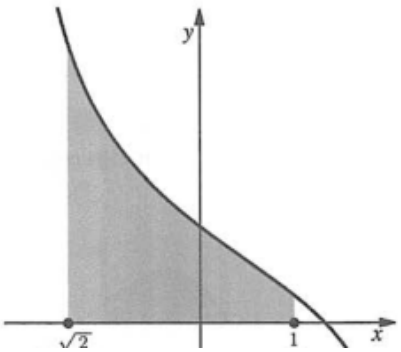
$$f(x) = \frac{5x+20}{\sqrt{25-x^2}}, \qquad |x| < 5.$$

Find the exact mean value of $f(x)$ over the interval $\left[-\frac{5}{2}, 0\right]$.

2.25 — Question 28. The curve shown in the diagram below has equation

$$y = \frac{4-9x}{\sqrt{4-9x^2}}.$$

Find the exact area of the shaded region.



2.25 — Question 29. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{1}{1+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{1-4x^2}}\,dx$

d) $\displaystyle\int \frac{1}{1+9x^2}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{9-x^2}}\,dx$

f) $\displaystyle\int \frac{1}{4+x^2}\,dx$

g) $\displaystyle\int \frac{1}{\sqrt{1-25x^2}}\,dx$

h) $\displaystyle\int \frac{1}{1+81x^2}\,dx$

i) $\displaystyle\int \frac{1}{\sqrt{4-9x^2}}\,dx$

j) $\displaystyle\int \frac{1}{9+4x^2}\,dx$

k) $\displaystyle\int \frac{1}{\sqrt{25-4x^2}}\,dx$

l) $\displaystyle\int \frac{1}{49+36x^2}\,dx$

m) $\displaystyle\int \frac{1}{\sqrt{16-3x^2}}\,dx$

n) $\displaystyle\int \frac{1}{4+5x^2}\,dx$

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2.25 — Question 30. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \sqrt{1-x^2}\,dx$

b) $\displaystyle\int (1-x^2)^{3/2}\,dx$

2.25 — Question 31. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \frac{1}{(1+x^2)^2}\,dx$

b) $\displaystyle\int \frac{1}{(4+x^2)^2}\,dx$

c) $\displaystyle\int \frac{1}{(1+x^2)^3}\,dx$

d) $\displaystyle\int \frac{1}{(9+4x^2)^3}\,dx$

2.25 — Question 32. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \sqrt{36-x^2}\,dx$

b) $\displaystyle\int (1-4x^2)^{3/2}\,dx$

2.25 — Question 33. Complete the square for each of these:

a) $26 - 10x + x^2$
b) $101 + 20x + x^2$
c) $2 + 4x + 4x^2$
d) $5 - 12x + 9x^2$
e) $25 - 8x + x^2$
f) $29 + 8x + 4x^2$

2.25 — Question 34.

a) Write $2 + 2x + x^2$ in the form $p + (x + q)^2$, where p and q are integers.
b) Hence, using the substitution $x + 1 = \tan u$, find $\displaystyle\int \frac{1}{2 + 2x + x^2}\, dx$.

2.25 — Question 35. Find:

a) $\displaystyle\int \frac{1}{10-6x+x^2}\,dx$

b) $\displaystyle\int \frac{1}{65+16x+x^2}\,dx$

2.25 — Question 36.

a) Write $10+18x+9x^2$ in the form $p+q(x+r)^2$, where p, q and r are integers.

b) Hence, using the substitution $3x+3=\tan u$, find $\displaystyle\int \frac{1}{10+18x+9x^2}\,dx$.

2.25 — Question 37. Find:

a) $\displaystyle\int \frac{1}{10+12x+4x^2}\,dx$

b) $\displaystyle\int \frac{1}{50-28x+4x^2}\,dx$

2.25 — Question 38.

a) Write $13-6x+x^2$ in the form $p+(x+q)^2$ where p and q are integers.

b) Hence, using the substitution $\frac{x-3}{2}=\tan u$, find $\displaystyle\int \frac{1}{13-6x+x^2}\,dx$.

2.25 — Question 39. Find:

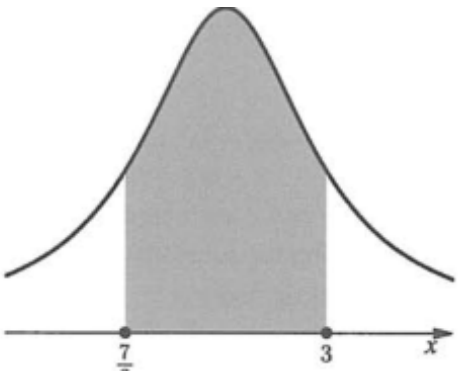
a) $\displaystyle\int \frac{1}{26+8x+2x^2}\,dx$

b) $\displaystyle\int \frac{6}{5+12x+8x^2}\,dx$

2.25 — Question 40. The curve shown in the diagram below has equation

$$y = \frac{2}{65-48x+9x^2}.$$

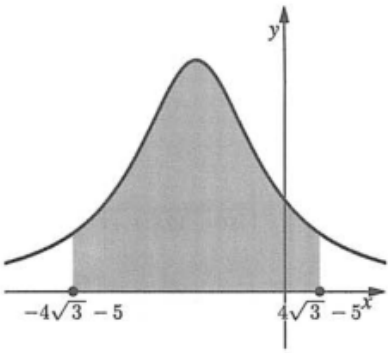
Find the exact area of the shaded region.



2.25 — Question 41. The curve shown in the diagram below has equation

$$y = \frac{7}{41 + 10x + x^2}.$$

Find the exact area of the shaded region.



2.25 — Question 42. Complete the square for each of these:

a) $-8 + 6x - x^2$
b) $-48 - 14x - x^2$
c) $-15 + 16x - 4x^2$
d) $-224 - 90x - 9x^2$
e) $-32 + 12x - x^2$
f) $-128 - 48x - 4x^2$

2.25 — Question 43.

a) Write $-63 + 16x - x^2$ in the form $p - (x + q)^2$ where p and q are integers.

b) Hence, using the substitution $x - 8 = \sin u$, find $\displaystyle\int \frac{1}{\sqrt{-63 + 16x - x^2}}\, dx$.

2.25 — Question 44. Find:

a) $\displaystyle\int \frac{1}{\sqrt{-3 - 4x - x^2}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{-80 + 18x - x^2}}\, dx$

2.25 — Question 45.

a) Write $-35 - 24x - 4x^2$ in the form $p - q(x + r)^2$ where p, q and r are integers.

b) Hence, using the substitution $2x + 6 = \sin u$, find $\displaystyle\int \frac{1}{\sqrt{-35 - 24x - 4x^2}}\, dx$.

2.25 — Question 46. Find:

a) $\displaystyle\int \frac{1}{\sqrt{-63 - 32x - 4x^2}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{-99 + 100x - 25x^2}}\, dx$

2.25 — Question 47.

a) Write $-27 + 24x - 4x^2$ in the form $p - q(x + r)^2$ where p, q and r are integers.

b) Hence, using the substitution $\frac{2}{3}x - 2 = \sin u$, find $\displaystyle\int \frac{1}{\sqrt{-27 + 24x - 4x^2}}\, dx$.

2.25 — Question 48. Find:

a) $\displaystyle\int \frac{1}{\sqrt{-11 - 36x - 9x^2}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{3 + 10x - 25x^2}}\, dx$

2.25 — Question 49. Show that

$$\int_2^{9/2} \frac{1}{\sqrt{21+4x-x^2}}\,dx = k\pi,$$

where k is a rational number to be found.

2.25 — Question 50. A curve has equation

$$y = \frac{1}{\sqrt{400+x^2}}.$$

Find the exact volume formed when the region between the curve, the x-axis, the y-axis and the line $x = 20$ is rotated through 2π radians about the x-axis.

2.25 — Question 51. Show that

$$\int_{2/5}^{\infty} \frac{1}{4+25x^2}\,dx = k\pi,$$

where k is a rational number to be found.

2.26 — Question 1. Show that

$$\frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{\sqrt{x^2+1}}.$$

2.26 — Question 2. Show that

$$\frac{d}{dx}(\operatorname{arcosh} x) = \frac{1}{\sqrt{x^2-1}}.$$

2.26 — Question 3. Show that

$$\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}.$$

2.26 — Question 4. Differentiate each of the following:

a) $y = \operatorname{arsinh} 3x$
b) $y = \operatorname{arcosh} 2x$
c) $y = \operatorname{artanh} 5x$
d) $y = \operatorname{arsinh} \frac{1}{2}x$
e) $y = \operatorname{arcosh} \frac{x}{4}$
f) $y = \operatorname{artanh} \left(-\frac{x}{6}\right)$

2.26 — Question 5. Differentiate each of the following:

a) $y = 2 \operatorname{arsinh} 2x - 2 \operatorname{arcosh} 4x$
b) $y = \frac{1}{3} \operatorname{artanh} 3x - \frac{2}{3} \operatorname{artanh} 6x$

2.26 — Question 6. Find the equation of the tangent to the curve $y = 4\operatorname{arsinh} x$ when $x = 2$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

2.26 — Question 7. Find the equation of the tangent to the curve $y = -6\operatorname{arcosh} \frac{x}{2}$ when $x = 4$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

2.26 — Question 8. Find the equation of the tangent to the curve $y = 10\operatorname{artanh}\frac{x}{5}$ when $x = 3$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

2.26 — Question 9. Determine whether the curve with equation

$$y = \operatorname{arsinh} 4x - \operatorname{arcosh} 2x$$

is convex (concave upwards) or concave (concave downwards) when $x = 2$.

2.26 — Question 10. It is given that the curve with equation

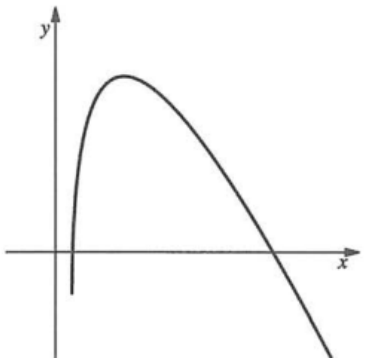
$$y = \sqrt{x^2 + 1} + \operatorname{arsinh} x$$

has a single stationary point. Find the exact coordinates of the stationary point.

2.26 — Question 11. The graph of $y = f(x)$, where

$$f(x) = 12 \operatorname{arcosh} x - 3x, \qquad x \geq 1,$$

is shown below. Find the exact range of f, giving your answer in set notation.



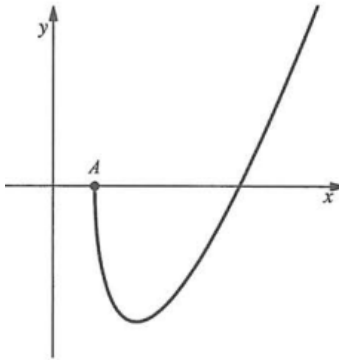
2.26 — Question 12. The graph of $y = f(x)$ is shown below, where

$$f(x) = 2\sqrt{x^2 - 9} - 12\,\text{arcosh}\,\frac{x}{3}.$$

The point A lies on the curve and on the x-axis.

a) State the coordinates of the point A.

b) Find the range of f in set notation.



2.26 — Question 13. It is given that the curve with equation

$$y = 4\,\text{artanh}\,x + \ln(x + 1) - 6x$$

has two stationary points. It is also given that the curve has a single point of inflection.

a) Find the exact coordinates of both stationary points.

b) Find the exact x-coordinate of the point of inflection. Fully justify your answer.

2.26 — Question 14. Differentiate each of the following:

a) $y = \text{arsinh}(\sqrt{x})$

b) $y = \text{arcosh}\left(\frac{1}{\sqrt{x}}\right)$

c) $y = \text{artanh}(\sqrt[3]{x})$

2.26 — Question 15. It is given that $x > 0$.

a) Show that $\dfrac{d}{dx}\left(\text{arsinh}\,\dfrac{1}{x}\right) = -\dfrac{1}{x\sqrt{x^2+1}}$.

b) A curve has equation $y = \sqrt{x^2+1} - \text{arsinh}\,\frac{1}{x}$. Show that the curve has no stationary points.

2.26 — Question 16.

a) Show that

$$\frac{d}{dx}\left(\operatorname{artanh}\frac{2x}{\sqrt{4x^2+1}}\right)=\frac{2}{\sqrt{4x^2+1}}.$$

b) A curve has equation

$$y=\sqrt{4x^2+1}-\operatorname{artanh}\frac{2x}{\sqrt{4x^2+1}}.$$

Show that the curve has a single stationary point and find its exact coordinates.

2.26 — Question 17. It is given that

$$f(x)=\operatorname{arcosh}2x+\operatorname{arcosh}\frac{2}{x}.$$

For $x>0$, find the exact coordinates of the stationary point of the curve $y=f(x)$.

2.26 — Question 18. Differentiate each of the following:

a) $y = \operatorname{arsinh}(\cosh x)$
b) $y = \operatorname{arcosh}(\tanh x)$
c) $y = \operatorname{artanh}(\sinh x)$
d) $y = \operatorname{arsinh}(\tanh x)$
e) $y = \operatorname{arcosh}(\sinh x)$
f) $y = \operatorname{artanh}(\cosh x)$

2.26 — Question 19. Differentiate each of the following:

a) $y = x^2 \operatorname{arsinh} x$
b) $y = \sinh x \operatorname{arsinh} x$
c) $y = \cosh x \operatorname{arcosh} x$
d) $y = \operatorname{arsinh} x \operatorname{arcosh} x$
e) $y = \operatorname{artanh} 2x \operatorname{artanh} 4x$

2.26 — Question 20. Differentiate each of the following:

a) $y = \dfrac{x}{\operatorname{arsinh} x}$

b) $y = \dfrac{\operatorname{arsinh} x}{\operatorname{arcosh} x}$

c) $y = \dfrac{\operatorname{arcosh} 3x}{\operatorname{arsinh} 6x}$

d) $y = \dfrac{\operatorname{arsinh} 4x}{\operatorname{artanh} 8x}$

2.26 — Question 21. It is given that

$$y = \sinh(2 \operatorname{arcosh} x), \qquad x \geq 1.$$

Show that

$$(x^2 - 1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 4y.$$

2.26 — Question 22. It is given that

$$y = (\operatorname{arsinh} 4x)^2, \qquad x > 0.$$

Show that

$$(16x^2 + 1)\frac{d^2y}{dx^2} + 16x\frac{dy}{dx} = 32.$$

2.26 — Question 23. Show that

$$\frac{d}{dx}(\operatorname{arcosech} x) = -\frac{1}{x\sqrt{x^2 + 1}}.$$

2.26 — Question 24. Show that

$$\frac{d}{dx}(\operatorname{arcsech} x) = -\frac{1}{x\sqrt{1-x^2}}.$$

2.26 — Question 25. Show that

$$\frac{d}{dx}(\operatorname{arcoth} x) = \frac{1}{1-x^2}.$$

2.26 — Question 26. A curve has equation

$$y = x^2 \operatorname{arcosech} 2x, \qquad x > 0.$$

a) Differentiate y with respect to x.
b) Hence find the exact gradient of the curve when $x = \frac{1}{2}$.

2.26 — Question 27. A curve has equation

$$y = \operatorname{arcsech} x^2 + \operatorname{arcoth} 3x,$$

where $-1 \leq x < -\frac{1}{3}$. Find the exact gradient of the curve when $x = -\frac{1}{2}$.

2.27 — Question 1. Use the substitution $x = \sinh u$ to show that

$$\int \frac{1}{\sqrt{x^2+1}}\,dx = \operatorname{arsinh} x + c.$$

2.27 — Question 2. Use the substitution $x = \cosh u$ to show that

$$\int \frac{1}{\sqrt{x^2-1}}\,dx = \operatorname{arcosh} x + c.$$

2.27 — Question 3. Use the substitution $x = \tanh u$ to show that

$$\int \frac{1}{1-x^2}\,dx = \operatorname{artanh} x + c.$$

2.27 — Question 4. Find:

a) $\displaystyle\int \frac{4}{\sqrt{x^2-1}}\,dx$

b) $\displaystyle\int \frac{1}{2\sqrt{x^2+1}}\,dx$

c) $\displaystyle\int \frac{6}{1-x^2}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{4x^2-1}}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{9x^2+1}}\,dx$

f) $\displaystyle\int \frac{1}{1-25x^2}\,dx$

g) $\displaystyle\int \frac{10}{\sqrt{16x^2-1}}\,dx$

h) $\displaystyle\int \frac{\sqrt{8}}{\sqrt{2x^2+1}}\,dx$

i) $\displaystyle\int \frac{100}{1-\frac{1}{5}x^2}\,dx$

2.27 — Question 5.

a) Show that

$$\frac{1}{15-x^2} \equiv \frac{1}{15(1-(kx)^2)},$$

where k is a real number to be found.

b) Hence find $\displaystyle\int \frac{1}{15-x^2}\,dx$.

2.27 — Question 6.

a) Show that

$$\frac{1}{\sqrt{x^2+3}} \equiv \frac{1}{\sqrt{3}\sqrt{(kx)^2+1}},$$

where k is a real number to be found.

b) Hence find $\displaystyle\int \frac{1}{\sqrt{x^2+3}}\,dx$.

2.27 — Question 7. Find:

a) $\int \frac{1}{25-x^2}\,dx$

b) $\int \frac{1}{\sqrt{x^2+9}}\,dx$

c) $\int \frac{1}{\sqrt{x^2-4}}\,dx$

d) $\int \frac{1}{8-x^2}\,dx$

e) $\int \frac{1}{\sqrt{x^2+12}}\,dx$

f) $\int \frac{1}{\sqrt{x^2-20}}\,dx$

g) $\int \frac{5}{10-x^2}\,dx$

h) $\int \frac{14}{\sqrt{x^2+7}}\,dx$

i) $\int \frac{18}{\sqrt{x^2-3}}\,dx$

j) $\int \frac{1}{9-4x^2}\,dx$

k) $\int \frac{1}{\sqrt{9x^2+4}}\,dx$

l) $\int \frac{1}{\sqrt{25x^2-16}}\,dx$

m) $\int \frac{15}{4-\frac{1}{4}x^2}\,dx$

n) $\int \frac{3}{\sqrt{16x^2+\frac{1}{9}}}\,dx$

o) $\int \frac{18}{\ldots}$

2.27 — Question 8. Given that a and k are constants, find:

a) $\int \frac{1}{a^2-x^2}\,dx$

b) $\int \frac{1}{\sqrt{x^2+a^2}}\,dx$

c) $\int \frac{1}{\sqrt{k^2x^2-a^2}}\,dx$

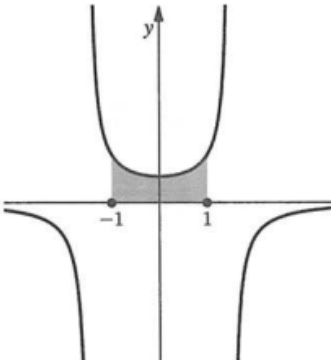
2.27 — Question 9. Find the exact values of:

a) $\displaystyle\int_{-1/2}^{1/2} \frac{1}{1-x^2}\,dx$

b) $\displaystyle\int_{1}^{2} \frac{1}{\sqrt{4x^2-1}}\,dx$

c) $\displaystyle\int_{-5}^{5} \frac{30}{\sqrt{36x^2+25}}\,dx$

2.27 — Question 10. The diagram below shows the curve $y = \frac{3}{9-4x^2}$. Show that the exact shaded area is $\frac{1}{2}\ln 5$.



2.27 — Question 11. Show that

$$\int_{3/5}^{4/5} \left(\frac{5}{\sqrt{25x^2-4}} + \frac{31}{155x-92} \right) dx = \ln(6 - 2\sqrt{5} + 3\sqrt{3} - \sqrt{15}).$$

2.27 — Question 12. Show that

$$\int_{1}^{3/2} \frac{4}{\sqrt{4x^2-1}}\, dx = \ln(a + b\sqrt{6} + c\sqrt{3} + d\sqrt{2}),$$

where a, b, c and d are integers to be found.

2.27 — Question 13. It is given, for $x \in \mathbb{R}$, that

$$f(x) = \frac{19 - 39x^2}{(1 - x^2)(4 - 9x^2)}.$$

a) Write $f(x)$ in the form $\dfrac{Ax + B}{1 - x^2} + \dfrac{Cx + D}{4 - 9x^2}$, where A, B, C and D are constants to be found.

b) Show that $\displaystyle\int f(x)\,dx = p\,\text{artanh}(qx) + r\,\text{artanh}(sx) + c$, where p, q, r and s are constants to be found, and c is a constant of integration.

2.27 — Question 14. Find:

a) $\displaystyle\int \frac{12x^2 + 1}{(1 - 9x^2)(1 - 16x^2)}\,dx$

b) $\displaystyle\int \frac{10x^2 + 47}{(1 - 4x^2)(25 - x^2)}\,dx$

c) $\displaystyle\int \frac{108x^4 + 12x^3 - 237x^2 - 27x + 25}{(1 - 9x^2)(9 - 4x^2)}\,dx$

2.27 — Question 15. Given that

$$\frac{1-2x}{1-x^2} \equiv \frac{1}{1-x^2} - \frac{2x}{1-x^2},$$

find $\displaystyle\int \frac{1-2x}{1-x^2}\,dx$.

2.27 — Question 16. Find:

a) $\displaystyle\int \frac{6x+1}{1-9x^2}\,dx$

b) $\displaystyle\int \frac{2-8x}{4-x^2}\,dx$

c) $\displaystyle\int \frac{8-3x}{16-9x^2}\,dx$

d) $\displaystyle\int \frac{10x-4}{25-100x^2}\,dx$

2.27 — Question 17. Given

$$\frac{1+x}{\sqrt{x^2-1}} \equiv \frac{1}{\sqrt{x^2-1}} + \frac{x}{\sqrt{x^2-1}},$$

find $\displaystyle\int \frac{1+x}{\sqrt{x^2-1}}\,dx$.

2.27 — Question 18. Given

$$\frac{4x+1}{\sqrt{x^2+1}} \equiv \frac{4x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}},$$

find $\displaystyle\int \frac{4x+1}{\sqrt{x^2+1}}\,dx$.

2.27 — Question 19. Find:

a) $\displaystyle\int \frac{2x-1}{\sqrt{4x^2-1}}\,dx$

b) $\displaystyle\int \frac{1-3x}{\sqrt{9x^2+1}}\,dx$

c) $\displaystyle\int \frac{4+6x}{\sqrt{9x^2-4}}\,dx$

d) $\displaystyle\int \frac{8x-3}{\sqrt{4x^2+36}}\,dx$

2.27 — Question 20. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\displaystyle\int \frac{1}{\sqrt{x^2+1}}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{x^2-1}}\,dx$

c) $\displaystyle\int \frac{1}{1-x^2}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{4x^2+1}}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{x^2-9}}\,dx$

f) $\displaystyle\int \frac{1}{1-9x^2}\,dx$

g) $\displaystyle\int \frac{1}{\sqrt{x^2+25}}\,dx$

h) $\displaystyle\int \frac{1}{\sqrt{36x^2-1}}\,dx$

i) $\displaystyle\int \frac{1}{100-x^2}\,dx$

j) $\displaystyle\int \frac{1}{\sqrt{49x^2+1}}\,dx$

k) $\displaystyle\int \frac{1}{\sqrt{x^2-64}}\,dx$

l) $\displaystyle\int \frac{1}{1-144x^2}\,dx$

m) $\displaystyle\int \frac{1}{\sqrt{4x^2+9}}\,dx$

n) $\displaystyle\int \frac{1}{\sqrt{9x^2-16}}\,dx$

o) $\displaystyle\int \frac{1}{4-25x^2}\,dx$

p) $\displaystyle\int \frac{1}{\sqrt{36x^2+25}}\,dx$

2.27 — Question 21. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\int \sqrt{x^2 - 1}\, dx$

b) $\int (x^2 + 1)^{3/2}\, dx$

2.27 — Question 22. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\int \sqrt{x^2 - 1}\, dx$

b) $\int (x^2 - 1)^{3/2}\, dx$

2.27 — Question 23. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\displaystyle\int \frac{1}{(1-x^2)^2}\,dx$

b) $\displaystyle\int \frac{1}{(1-9x^2)^2}\,dx$

c) $\displaystyle\int \frac{1}{(1-x^2)^3}\,dx$

d) $\displaystyle\int \frac{1}{(4-x^2)^3}\,dx$

2.27 — Question 24. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\displaystyle\int \sqrt{25x^2+1}\,dx$

b) $\displaystyle\int (x^2+9)^{3/2}\,dx$

c) $\displaystyle\int \sqrt{x^2-36}\,dx$

d) $\displaystyle\int (81x^2-1)^{3/2}\,dx$

2.27 — Question 25. Complete the square for each of these:

a) $x^2 + 4x + 3$
b) $x^2 - \frac{10}{3}x + \frac{34}{9}$
c) $4x^2 + 24x + 37$
d) $9x^2 - 18x + 8$
e) $x^2 + 6x + 18$
f) $9x^2 - 36x + 32$

2.27 — Question 26.

a) Write $x^2 + 20x + 101$ in the form $(x + p)^2 + q$, where p and q are integers.
b) Hence, using the substitution $x + 10 = \sinh u$, find $\displaystyle\int \frac{1}{\sqrt{x^2 + 20x + 101}}\, dx$.

2.27 — Question 27.

a) Write $x^2 - 14x + 48$ in the form $(x+p)^2 + q$, where p and q are integers.

b) Hence, using the substitution $x - 7 = \cosh u$, find $\displaystyle\int \frac{1}{\sqrt{x^2 - 14x + 48}}\, dx$.

2.27 — Question 28. Find:

a) $\displaystyle\int \frac{1}{\sqrt{x^2 - 10x + 26}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{x^2 + \frac{2}{3}x - \frac{8}{9}}}\, dx$

2.27 — Question 29. Find the exact mean value of $f(x)$ over the interval $\left[-\frac{2}{5}, \frac{2}{5}\right]$, given

$$f(x) = \frac{1}{\sqrt{x^2 - \frac{4}{5}x + \frac{29}{25}}}, \qquad x \in \mathbb{R}.$$

2.27 — Question 30.

a) Write $4x^2 - 40x + 101$ in the form $p(x+q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $2x - 10 = \sinh u$, find $\displaystyle\int \frac{1}{\sqrt{4x^2 - 40x + 101}}\, dx$.

2.27 — Question 31.

a) Write $16x^2 + 96x + 143$ in the form $p(x + q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $4x + 12 = \cosh u$, find $\displaystyle\int \frac{1}{\sqrt{16x^2 + 96x + 143}}\, dx$.

2.27 — Question 32. Find:

a) $\displaystyle\int \frac{1}{\sqrt{9x^2 - 144x + 575}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{2x^2 + 12x + 19}}\, dx$

2.27 — Question 33.

a) Write $4x^2 - 56x + 205$ in the form $p(x+q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $\frac{2}{3}(x-7) = \sinh u$, find $\displaystyle\int \frac{1}{\sqrt{4x^2 - 56x + 205}}\, dx$.

2.27 — Question 34.

a) Write $3x^2 + 6x - 1$ in the form $p(x+q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $\frac{\sqrt{3}}{2}(x+1) = \cosh u$, find $\displaystyle\int \frac{1}{\sqrt{3x^2 + 6x - 1}}\, dx$.

2.27 — Question 35. Find:

a) $\displaystyle\int \frac{1}{\sqrt{25x^2 - 100x + 91}}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{5x^2 + 10\sqrt{5}x + 29}}\,dx$

2.27 — Question 36. Show that

$$\int_{-1}^{1} \frac{1}{\sqrt{4x^2 + 8x + 13}}\,dx = \frac{1}{2}\ln k,$$

where k is a positive integer to be found.

2.27 — Question 37. Complete the square for each of these:

a) $-3 - 4x - x^2$

b) $-24 + 10x - x^2$

c) $-15 - 16x - 4x^2$

d) $-143 + 72x - 9x^2$

e) $-45 - 14x - x^2$

f) $8 + 4x - 4x^2$

2.27 — Question 38.

a) Write $-63 + 16x - x^2$ in the form $p - (x + q)^2$, where p and q are integers.

b) Hence, using the substitution $x - 8 = \tanh u$, find $\displaystyle\int \frac{1}{(-63 + 16x - x^2)}\, dx$.

2.27 — Question 39. Find:

a) $\displaystyle\int \frac{1}{(-99-20x-x^2)}\,dx$

b) $\displaystyle\int \frac{1}{(-168+26x-x^2)}\,dx$

2.27 — Question 40.

a) Write $-35+24x-4x^2$ in the form $p+q(x+r)^2$, where p, q and r are integers.

b) Hence, using the substitution $2x-6=\tanh u$, find $\displaystyle\int \frac{1}{(-35+24x-4x^2)}\,dx$.

2.27 — Question 41. Find:

a) $\displaystyle\int \frac{1}{(-224-90x-9x^2)}\,dx$

b) $\displaystyle\int \frac{1}{(-24+50x-25x^2)}\,dx$

2.27 — Question 42.

a) Write $11+10x-x^2$ in the form $p-(x+q)^2$, where p and q are integers.

b) Hence, using the substitution $\frac{x-5}{6}=\tanh u$, find $\displaystyle\int \frac{1}{(11+10x-x^2)}\,dx$.

2.27 — Question 43. Find:

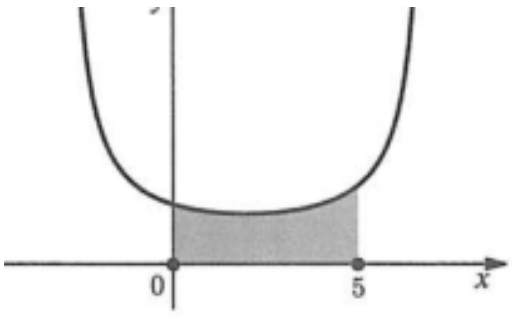
a) $\int \frac{1}{(91-18x-9x^2)}\,dx$

b) $\int \frac{1}{(-20-24x-4x^2)}\,dx$

2.27 — Question 44. The curve shown in the diagram below has equation

$$y = \frac{5}{63+12x-3x^2}.$$

Find the exact area of the shaded region.



2.27 — Question 45. A curve has equation

$$y = \frac{2}{\sqrt{-16 + 16x - 3x^2}}.$$

Find the exact volume formed when the region between the curve, the x-axis, and the lines $x = 2$ and $x = 3$ is rotated through 2π radians about the x-axis.

2.27 — Question 46. Use the substitution $x = \cosh u$ to find $\displaystyle\int \sqrt{x^2 - 1}\, dx$.

2.27 — Question 47. Use the substitution $x = \sinh u$ to find $\int \sqrt{x^2 + 1}\, dx$.

2.27 — Question 48. Use a suitable substitution to find $\int \sqrt{9x^2 - 1}\, dx$.

2.27 — Question 49. Find the exact value of

$$\int_{12}^{13} \sqrt{x^2 - 144}\, dx.$$

2.27 — Question 50. Show that

$$\int_0^2 \sqrt{25x^2 + 1}\, dx = \sqrt{101} + \frac{1}{10}\ln(10 + \sqrt{101}).$$

2.27 — Question 51. Find:

a) $\int \sqrt{x^2 - 10x + 26}\, dx$

b) $\int \sqrt{x^2 + 6x + 8}\, dx$

2.27 — Question 52. Show that

$$\int_3^5 \sqrt{x^2 - 8x + 17}\, dx = \sqrt{2} + \ln(1 + \sqrt{2}).$$

2.27 — Question 53. Use a suitable substitution to find $\int \frac{x^2}{\sqrt{x^2-1}}\,dx$.

2.27 — Question 54. Use a suitable substitution to find $\int \frac{x^2}{\sqrt{x^2+1}}\,dx$.

2.27 — Question 55. Use a suitable substitution to find $\displaystyle\int \frac{x^3}{\sqrt{x^2+1}}\,dx$.

2.27 — Question 56. Use the substitution $x = \sinh^2 u$ to find $\displaystyle\int \sqrt{\frac{x}{x+1}}\,dx$.

2.27 — Question 57. Use the substitution $x = \cosh u$ to find $\int (x^2 - 1)^{3/2}\, dx$.

2.28 — Question 1. For each of the following, suggest a substitution for x that reduces the integral to $\int C\, d\theta$, where C is a constant.

a) $\displaystyle\int \frac{1}{\sqrt{a^2 - x^2}}\, dx$, for constant a

b) $\displaystyle\int \frac{1}{\sqrt{a^2 + x^2}}\, dx$, for constant a

c) $\displaystyle\int \frac{1}{a^2 - x^2}\, dx$, for constant a

d) $\displaystyle\int \frac{1}{\sqrt{x^2 - a^2}}\, dx$, for constant a

e) $\displaystyle\int \frac{1}{a^2 + x^2}\, dx$, for constant a

2.28 — Question 2. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\displaystyle\int \frac{1}{\sqrt{9x^2+400}}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{9-16x^2}}\,dx$

c) $\displaystyle\int \frac{1}{36-x^2}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{625x^2+1}}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{121x^2-16}}\,dx$

f) $\displaystyle\int \frac{1}{1+25x^2}\,dx$

g) $\displaystyle\int \frac{1}{\sqrt{100x^2+9}}\,dx$

h) $\displaystyle\int \frac{1}{\sqrt{4-121x^2}}\,dx$

i) $\displaystyle\int \frac{16}{\sqrt{x^2-4}}\,dx$

j) $\displaystyle\int \frac{7}{144+49x^2}\,dx$

k) $\displaystyle\int \frac{24}{25-9x^2}\,dx$

l) $\displaystyle\int \frac{32}{\sqrt{81-2x^2}}\,dx$

m) $\displaystyle\int \frac{90}{\sqrt{9x^2-10}}\,dx$

n) $\displaystyle\int \frac{55}{50-121x^2}\,dx$

o) $\displaystyle\int \frac{36}{5+\ldots}\,dx$

2.28 — Question 3. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\displaystyle\int \frac{3}{\sqrt{9x^2+4}}\,dx$

b) $\displaystyle\int \frac{14}{4+49x^2}\,dx$

c) $\displaystyle\int \frac{144}{9-16x^2}\,dx$

d) $\displaystyle\int \frac{8}{\sqrt{16x^2-9}}\,dx$

e) $\displaystyle\int \sqrt{25-4x^2}\,dx$

f) $\displaystyle\int \frac{24576}{(16-9x^2)^3}\,dx$

g) $\displaystyle\int (4-49x^2)^{3/2}\,dx$

h) $\displaystyle\int \sqrt{36x^2+25}\,dx$

i) $\displaystyle\int \frac{900000}{(100+9x^2)^3}\,dx$

j) $\displaystyle\int \frac{280}{(25-49x^2)^2}\,dx$

k) $\displaystyle\int \frac{250}{(25+2x^2)^2}\,dx$

l) $\displaystyle\int (3x^2-4)^{3/2}\,dx$

2.28 — Question 4. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\int \frac{1}{\sqrt{1-(3x-1)^2}}\,dx$

b) $\int \frac{1}{\sqrt{(x-1)^2+9}}\,dx$

c) $\int \frac{1}{\sqrt{3-(2x+5)^2}}\,dx$

d) $\int \frac{1}{\sqrt{(3x-2)^2+8}}\,dx$

e) $\int \frac{1}{8+(4x-1)^2}\,dx$

f) $\int \frac{4}{11-(5x+3)^2}\,dx$

2.28 — Question 5. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\int \frac{1}{4x^2+12x+10}\,dx$

b) $\int \frac{3}{\sqrt{3-4x-4x^2}}\,dx$

c) $\int \frac{2}{x^2-6x+14}\,dx$

d) $\int \frac{1}{\sqrt{7+4x-x^2}}\,dx$

e) $\int \frac{4}{3x^2+8x+6}\,dx$

f) $\int \frac{7}{\sqrt{5x^2+2x+12}}\,dx$

2.28 — Question 6. By first suggesting a suitable substitution, show that:

a) $\displaystyle\int_0^1 \frac{2}{\sqrt{5-(2x-1)^2}}\,dx = 2\arcsin\left(\frac{\sqrt{5}}{5}\right)$

b) $\displaystyle\int_0^2 \frac{8}{\sqrt{4+(x-1)^2}}\,dx = 8\ln\left(\frac{3+\sqrt{5}}{2}\right)$

c) $\displaystyle\int_{-3}^{-3/2} \frac{3}{45+24x+4x^2}\,dx = \frac{\pi}{8}$

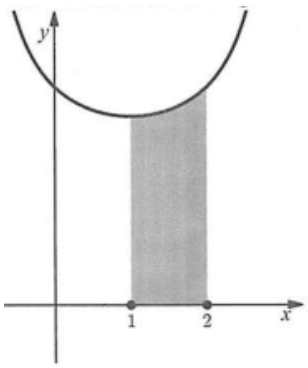
d) $\displaystyle\int_0^{3/2} \frac{2}{\sqrt{4x^2-12x+15}}\,dx = \ln\left(\frac{\sqrt{2}}{\sqrt{5}-\sqrt{3}}\right)$

e) $\displaystyle\int_{-1}^4 \sqrt{x^2+2x+10}\,dx = \frac{135}{32}+\frac{9}{2}\ln 2$

2.28 — Question 7. The curve shown in the diagram below has equation

$$y = \frac{5}{\sqrt{3+2x-x^2}}.$$

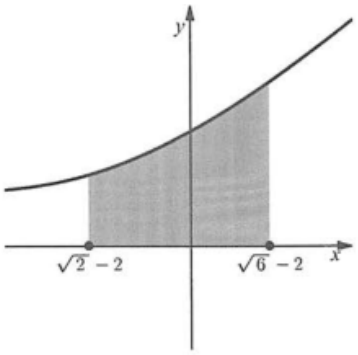
Find the exact area of the shaded region.



2.28 — Question 8. The curve shown in the diagram below has equation

$$y = x + \frac{4}{6 + 4x + x^2}.$$

Find the exact area of the shaded region.



2.29 — Question 1.

a) Use the substitution $x = \tan u$ to find $\displaystyle\int \frac{x^2}{1 + x^2}\, dx$.

b) By writing $\dfrac{x^2}{1 + x^2}$ in the form $A + \dfrac{B}{1 + x^2}$, where A and B are integers, confirm your result from part a).

2.29 — Question 2. Use the substitution $x = \tan u$, or otherwise, to find each of these.

a) $\displaystyle\int \frac{2x^2 - 1}{1 + x^2}\, dx$

b) $\displaystyle\int \frac{5x^2 + 9}{1 + x^2}\, dx$

2.29 — Question 3. Find $\displaystyle\int \frac{1}{\sqrt{x(1+x)}}\, dx$ by:

a) using the substitution $\sqrt{x} = \tan u$

b) using the substitution $\sqrt{x} = u$

2.29 — Question 4. Find $\displaystyle\int \frac{1}{(1+x^2)^{3/2}}\,dx$.

2.29 — Question 5. Use the substitution $u = \sqrt{x^2-1}$ to find $\displaystyle\int \frac{1}{x\sqrt{x^2-1}}\,dx$.

2.29 — Question 6.

a) Write $\sin^2\theta$ in terms of $\cos 2\theta$.

b) Use the substitution $x = \sin u$ to find $\displaystyle\int \frac{x^2}{\sqrt{1-x^2}}\,dx$.

2.29 — Question 7. Use a suitable substitution to find each of these.

a) $\displaystyle\int \frac{x^2}{\sqrt{1-4x^2}}\,dx$

b) $\displaystyle\int \frac{x^2}{\sqrt{4-x^2}}\,dx$

c) $\displaystyle\int \frac{x^2}{\sqrt{4-2x^2}}\,dx$

2.29 — Question 8. Use a suitable substitution to find each of these.

a) $\displaystyle\int \frac{2x^2+3}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{3-x^2}{\sqrt{1-9x^2}}\,dx$

c) $\displaystyle\int \frac{3x^2+1}{\sqrt{1-3x^2}}\,dx$

2.29 — Question 9. Find the exact value of $\displaystyle\int_{-1/4}^{1/4} \frac{8+8x^2}{\sqrt{1-4x^2}}\,dx$.

2.29 — Question 10.

a) Show that $\dfrac{d}{dx}\left(3x\sqrt{1-9x^2} - \arccos(3x)\right) = k\sqrt{1-9x^2}$, where k is a positive integer to be found.

b) Hence find the exact value of $\displaystyle\int_{1/6}^{\sqrt{3}/6} \sqrt{1-9x^2}\,dx$.

2.29 — Question 11.

a) Show that

$$\frac{d}{dx}\left(2(x-5)\sqrt{11+10x-x^2} + 72\arcsin\left(\frac{x-5}{6}\right)\right) = k\sqrt{11+10x-x^2},$$

where k is a positive integer to be found.

b) Hence find the exact value of $\displaystyle\int_{5}^{8} \sqrt{11+10x-x^2}\,dx$.

2.29 — Question 12.

a) Write $19 + 18x - x^2$ in the form $p - (x + q)^2$, where p and q are integers.

b) Using your answer to part a), use the substitution $\dfrac{x+q}{\sqrt{p}} = \sin u$ to show that

$$\int \sqrt{19 + 18x - x^2}\, dx = \frac{1}{2}(x - 9)\sqrt{19 + 18x - x^2} + 50 \arcsin\left(\frac{x-9}{10}\right) + c.$$

2.29 — Question 13. Use a suitable substitution to find $\displaystyle\int \sqrt{3 + 2x - x^2}\, dx$.

2.29 — Question 14.

a) Show that $\sin 3A \equiv 3\sin A - 4\sin^3 A$.

b) Show that $\cos 3A \equiv 4\cos^3 A - 3\cos A$.

c) Use a suitable substitution to find $\displaystyle\int \frac{6x^3 - 4x + 1}{\sqrt{1 - x^2}}\, dx$.

2.29 — Question 15.

a) Find $\dfrac{d}{dx}\left(\arctan\sqrt{\dfrac{x}{3}}\right)$.

b) Hence show that $\displaystyle\int_1^3 \frac{\sqrt{x}}{3x + x^2}\, dx = \frac{\pi}{6\sqrt{3}}$.

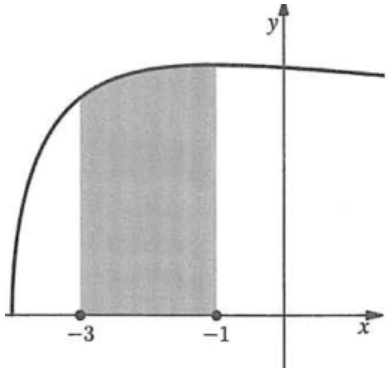
2.29 — Question 16. Use the substitution $u^2 = x - 3$, where $u > 0$, to find $\displaystyle\int \frac{\sqrt{x-3}}{x+6}\,dx$.

2.29 — Question 17. Use the substitution $u^2 = x + 1$, where $u > 0$, to find $\displaystyle\int \frac{x}{\sqrt{x+1}(x+10)}\,dx$.

2.29 — Question 18. The curve shown in the diagram below has equation

$$y = \frac{\sqrt{x+4}}{x+7}, \qquad x \geq -4.$$

The shaded region is the region bounded between the curve and the x-axis, and the lines $x = -3$ and $x = -1$. Show that the exact area of the shaded region is $\left(2\sqrt{3} - 2 - \frac{\pi}{2\sqrt{3}}\right)$ units^2.



2.29 — Question 19. Use integration by parts to find:

a) $\int \arcsin x \, dx$

b) $\int \arccos x \, dx$

c) $\int \arctan x \, dx$

2.29 — Question 20. Show that $\int_0^1 8\arctan x\,dx = 2\pi - \ln p$, where p is a positive integer to be found.

2.29 — Question 21. Find:

a) $\int \arcsin 2x\,dx$

b) $\int \arccos 3x\,dx$

c) $\int \arctan 4x\,dx$

2.29 — Question 22. Find:

a) $\int 8x \arcsin(x^2)\,dx$

b) $\int e^{2x} \arccos(e^{2x})\,dx$

c) $\int \sin x \arctan(\cos x)\,dx$

d) $\int \sec^2 x \arcsin(\tan x)\,dx$

e) $\int 3^x \arctan(3^x)\,dx$

2.29 — Question 23. Find:

a) $\int x \arcsin x\,dx$

b) $\int x^2 \arccos x\,dx$

c) $\int x^3 \arctan x\,dx$

2.29 — Question 24.

a) Write $\dfrac{10x^2}{1+x^2}$ in the form $A + \dfrac{B}{1+x^2}$.

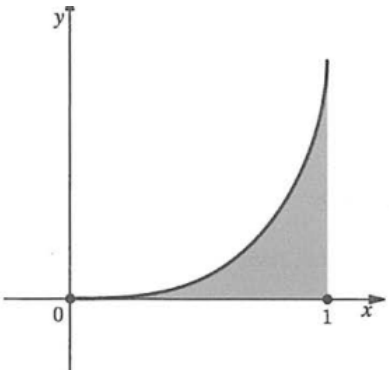
b) Find $\displaystyle\int 20x \arctan x \, dx$.

2.29 — Question 25.

a) Write $\dfrac{15x^2}{4+x^2}$ in the form $A + \dfrac{B}{4+x^2}$.

b) Find $\displaystyle\int 30x \arctan\left(\frac{x}{2}\right) \, dx$.

2.29 — Question 26. The curve shown in the diagram below has equation $y = 6x \arcsin(x^2)$, $0 \leq x \leq 1$. Find the exact area of the shaded region.



2.29 — Question 27. Find:

a) $\displaystyle\int \frac{\arcsin x}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{\arccos x}{\sqrt{1-x^2}}\,dx$

c) $\displaystyle\int \frac{\arctan x}{1+x^2}\,dx$

2.29 — Question 28. Show that $\displaystyle\int_{1/2}^{1/\sqrt{2}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}}\,dx = k\pi^3$, where k is a rational number to be found.

2.29 — Question 29. Find $\displaystyle\int \arctan x \ln(1+x^2)\,dx$.

2.29 — Question 30. Find:

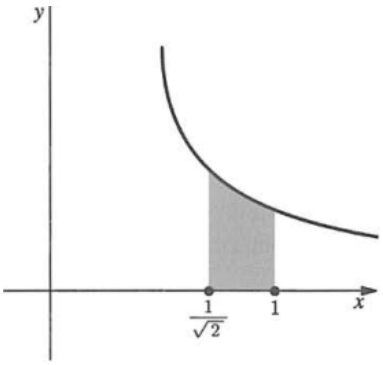
a) $\int \operatorname{arccosec} x \, dx, \quad x > 0$

b) $\int \operatorname{arcsec} x \, dx, \quad x > 0$

c) $\int \operatorname{arccot} x \, dx$

2.29 — Question 31. The curve shown in the diagram below has equation $y = \operatorname{arccosec} 2x$ for $x > \frac{1}{2}$. The shaded region is the region bounded between the curve and the x-axis, and the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$. Show that the exact area of the shaded region is

$$\left(\frac{1}{2}\ln\left(\frac{2+\sqrt{3}}{1+\sqrt{2}}\right) + \frac{\pi}{6} - \frac{\pi}{4\sqrt{2}}\right) \text{ units}^2.$$



2.30 — Question 1.

a) Show that $\operatorname{cosech} x = \dfrac{2e^x}{e^{2x} - 1}$.

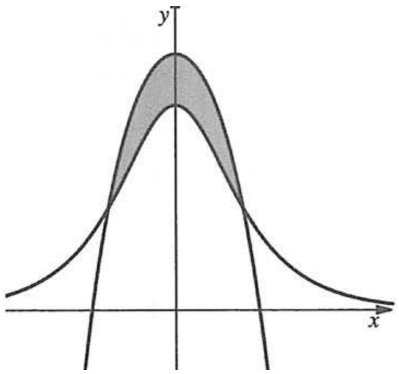
b) By using the substitution $u = e^x$, show that $\displaystyle\int \operatorname{cosech} x \, dx = -2 \operatorname{artanh}(e^x) + c$.

2.30 — Question 2.

a) Show that $\operatorname{sech} x = \dfrac{2e^x}{e^{2x} + 1}$.

b) By using the substitution $u = e^x$, show that $\displaystyle\int \operatorname{sech} x \, dx = 2 \arctan(e^x) + c$.

2.30 — Question 3. The shaded region shown in the diagram below is enclosed between the graphs of $y = 12\,\text{sech}\,x$ and $y = 24 - 9\cosh x$. Find the exact shaded area in the form $p\ln(7+4\sqrt{3}) + q\sqrt{3} + r\pi$, where p, q and r are integers.



2.30 — Question 4. Find:

a) $\displaystyle\int \tanh x\,dx$

b) $\displaystyle\int \coth x\,dx$

2.30 — Question 5. Find:

a) $\int \operatorname{cosech} x \coth x \, dx$

b) $\int \operatorname{sech} x \tanh x \, dx$

2.30 — Question 6.

a) By expressing $\sinh x$ and $\cosh x$ in terms of exponentials, prove that $\cosh^2 x - \sinh^2 x = 1$.

b) Find $\int \operatorname{cosech}^2 x \, dx$.

2.30 — Question 7. Find the equation of the curve y given that its gradient function is:

a) $\dfrac{dy}{dx} = \operatorname{cosech} 3x \coth 3x$

b) $\dfrac{dy}{dx} = 3\operatorname{cosech}^2\left(\dfrac{x}{9}\right)$

c) $\dfrac{dy}{dx} = 6\operatorname{sech} 9x \tanh 9x$

d) $\dfrac{dy}{dx} = 10\coth 2x$

2.30 — Question 8. Find $\displaystyle\int \frac{\cosh x}{\sinh^2 x + 1}\, dx$.

2.30 — Question 9. Find $\displaystyle\int \frac{\cos x}{\sqrt{\sin^2 x + 1}}\, dx$.

2.30 — Question 10.

a) Given that $u = \cosh^2 x$, show that $\dfrac{du}{dx} = \sinh 2x$.

b) Hence show that $\displaystyle\int \frac{\sinh 2x}{\cosh^4 x + 1}\, dx = \arctan(\cosh^2 x) + c$.

2.30 — Question 11. Use integration by parts to find:

a) $\int \operatorname{arsinh} x \, dx$

b) $\int \operatorname{arcosh} x \, dx$

c) $\int \operatorname{artanh} x \, dx$

2.30 — Question 12. Show that $\int_0^{1/3} 6 \operatorname{artanh} x \, dx = \ln\left(\frac{2^p}{3^q}\right)$, where p and q are positive integers.

2.30 — Question 13. Show that $\displaystyle\int_0^{\frac{\tanh 4}{2}} 4\,\operatorname{artanh}(2x)\,dx = 8\tanh 4 + \ln|1-\tanh^2 4|$.

2.30 — Question 14.

a) Find $\dfrac{d}{dx}(x\,\operatorname{arcosh} 3x)$.

b) Hence find $\displaystyle\int \operatorname{arcosh} 3x\,dx$.

2.30 — Question 15. Find $\int x^2 \operatorname{arcosh} x \, dx$.

2.30 — Question 16.

a) Find the values of A and B such that $\dfrac{2x^2}{1-x^2} = A + \dfrac{B}{1-x^2}$.

b) Hence find $\int 4x \operatorname{artanh} x \, dx$.

2.30 — Question 17. Find:

a) $\displaystyle\int \frac{\operatorname{arsinh} x}{\sqrt{x^2+1}}\,dx$

b) $\displaystyle\int \frac{\operatorname{arcosh} x}{\sqrt{x^2-1}}\,dx$

c) $\displaystyle\int \frac{\operatorname{artanh} x}{1-x^2}\,dx$

2.30 — Question 18. Find the exact value of the following integral in terms of logarithms:

$$\int_0^2 \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,dx.$$

2.30 — Question 19. Find the exact value of the following integral in terms of logarithms:

$$\int_{1/3}^{4} \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,dx.$$

2.30 — Question 20.

a) Show that $\dfrac{10}{5\cosh x - 4\sinh x}$ can be written in the form $\dfrac{pe^x}{e^{2x}+q}$, where p and q are integers.

b) Hence show that $\displaystyle\int_0^{\ln 3} \frac{30}{5\cosh x - 4\sinh x}\,dx = 5\pi - 20\arctan\frac{1}{3}$.

2.30 — Question 21. Use the substitution $u = \operatorname{artanh} x$ to find $\displaystyle\int \frac{1}{(1-x^2)((\operatorname{artanh} x)^2+1)}\,dx$.

2.30 — Question 22. A curve has equation $y = \operatorname{arcosh}\left(\frac{\sqrt{x}}{4}\right)$ for $x \geq \frac{1}{4}$.

a) Find $\dfrac{dy}{dx}$ and write it in a simplified form.

b) Hence show that $\displaystyle\int_{64}^{144} \frac{1}{\sqrt{x(x-16)}}\,dx = p\ln\left(\frac{q+p\sqrt{p}}{p+\sqrt{q}}\right)$, where p and q are positive integers.

2.30 — Question 23. Use integration by parts to find $\displaystyle\int \frac{x \operatorname{arsinh} x}{\sqrt{x^2+1}}\,dx$.

2.30 — Question 24.

a) Find $\dfrac{d}{dx}(\operatorname{arcosh} 2x)$.

b) Use integration by parts to find $\displaystyle\int \frac{x \operatorname{arcosh} 2x}{\sqrt{4x^2-1}}\,dx$.

2.30 — Question 25.

a) By using a double-angle identity, show that $\sinh(2\operatorname{arsinh} x) = 2x\sqrt{x^2+1}$.

b) Find $\displaystyle\int x \operatorname{arsinh} x \, dx$.

2.30 — Question 26.

a) By writing in exponential form, show that $\displaystyle\int \operatorname{cosech} x \, dx = \ln\left|\tanh\frac{x}{2}\right| + c$.

b) i) By writing in exponential form, show that $\displaystyle\int \operatorname{sech} x \, dx = 2\arctan(e^x) + c$.

ii) Show that $e^x = \dfrac{1+\tanh(x/2)}{1-\tanh(x/2)}$.

iii) Hence show that $\displaystyle\int \operatorname{sech} x \, dx$ can also be written as $2\arctan\left(\tanh\frac{x}{2}\right) + c$.

2.30 — Question 27.

a) By using exponential form, show that $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$.

b) Hence find $\int x \operatorname{sech} x \tanh x \, dx$.

2.31 — Question 1. Explain why each of the following are improper integrals.

a) $\int_1^\infty \sqrt{x} \, dx$

b) $\int_0^1 \frac{1}{3x^3} \, dx$

2.31 — Question 2. Showing the limiting process used, find:

a) $\displaystyle\int_2^\infty \frac{1}{x^2}\,dx$

b) $\displaystyle\int_2^\infty \frac{1}{x^3}\,dx$

c) $\displaystyle\int_2^\infty \frac{1}{x^4}\,dx$

2.31 — Question 3. Showing the limiting process used, find:

a) $\displaystyle\int_0^2 \frac{1}{x^2}\,dx$

b) $\displaystyle\int_0^2 \frac{1}{x^3}\,dx$

c) $\displaystyle\int_0^2 \frac{1}{x^4}\,dx$

2.31 — Question 4. Showing the limiting process used, find:

a) $\displaystyle\int_0^9 \frac{1}{\sqrt{x}}\,dx$

b) $\displaystyle\int_4^\infty \frac{1}{\sqrt{x}}\,dx$

2.31 — Question 5. Showing the limiting process used, find:

a) $\displaystyle\int_0^8 \frac{1}{x^{3/4}}\,dx$

b) $\displaystyle\int_8^\infty \frac{1}{x^{3/4}}\,dx$

2.31 — Question 6. Showing the limiting process used, find:

a) $\displaystyle\int_4^\infty 192x^{-4}\,dx$

b) $\displaystyle\int_4^\infty (192x^{-4}+1)\,dx$

c) $\displaystyle\int_4^\infty 192x^{-4}(x+1)\,dx$

2.31 — Question 7. Show clearly that $\displaystyle\int_0^4 \frac{1}{x^3\sqrt{x}}\,dx$ diverges.

2.31 — Question 8. Show that $\int_0^{1/64} \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt[3]{x}}\right) dx = \frac{5}{32}$.

2.31 — Question 9. Amelia is attempting to evaluate the integral $\int_{-2}^{2} \frac{35}{x^6}\, dx$. Amelia's working is shown below:

$$\int_{-2}^{2} \frac{35}{x^6}\, dx = \int_{-2}^{2} 35x^{-6}\, dx = [-7x^{-5}]_{-2}^{2} = ((-7 \times 2^{-5}) - (-7 \times (-2)^{-5})) = -\frac{7}{32} - \frac{7}{32} = -\frac{7}{16}.$$

a) Explain the error in Amelia's method and what she should do instead.

b) Show clearly that $\int_{-2}^{2} \frac{35}{x^6}\, dx$ diverges.

2.31 — Question 10. Find $\int_{-1}^{8} \frac{4}{x^{2/3}}\,dx$.

2.31 — Question 11. Find $\int_{-32}^{1} \frac{100}{x^{4/5}}\,dx$. **The following questions require A-Level Maths Integration methods, including the Integrals of Standard Functions, Reversing the Chain Rule, Integration by Substitution, Integration by Parts, and Partial Fractions.**

2.31 — Question 12. Explain why each of the following are improper integrals.

a) $\displaystyle\int_4^6 \frac{1}{x-5}\,dx$

b) $\displaystyle\int_0^1 \frac{e^x}{e^x-2}\,dx$

c) $\displaystyle\int_0^{2\pi/3} \frac{\sqrt{3}\sin x+\cos x}{\sin x-\sqrt{3}\cos x}\,dx$

d) $\displaystyle\int_0^1 x^4 \ln x\,dx$

2.31 — Question 13. Showing the limiting process used, find:

a) $\displaystyle\int_2^\infty \frac{1}{x}\,dx$

b) $\displaystyle\int_0^2 \frac{1}{x}\,dx$

2.31 — Question 14. Showing the limiting process used, find:

a) $\displaystyle\int_0^\infty \frac{1}{e^x}\,dx$

b) $\displaystyle\int_0^\infty \frac{1}{e^{2x}}\,dx$

2.31 — Question 15. Show that $\displaystyle\int_7^\infty (x-3)^{-5/2}\,dx = \frac{1}{12}$.

2.31 — Question 16. Showing the limiting process used, find $\int_{-2}^{62.5} \frac{2}{(2x+3)^6}\,dx$.

2.31 — Question 17. Showing the limiting process used, find:

a) $\int_{3}^{4} \frac{x}{x^2-9}\,dx$

b) $\int_{3}^{4} \frac{x}{\sqrt{x^2-9}}\,dx$

2.31 — Question 18. Showing the limiting process used, find:

a) $\displaystyle\int_0^{\infty} 4xe^{-x^2}\,dx$

b) $\displaystyle\int_{-\infty}^{\infty} 4xe^{-x^2}\,dx$

2.31 — Question 19. Showing the limiting process used, find $\displaystyle\int_{-\infty}^{0} \frac{e^x}{e^x+1}\,dx$.

2.31 — Question 20. Showing the limiting process used, find $\int_0^{\pi/2} \tan x \, dx$.

2.31 — Question 21. Showing the limiting process used, find $\int_{-1}^{2} \frac{x+1}{\sqrt{8-2x-x^2}} \, dx$.

2.31 — Question 22.

a) Show that $\lim_{a\to\infty}\left(\frac{6a-5}{3a+7}\right)=2$.

b) Showing the limiting process used, find $\int_4^\infty\left(\frac{6}{6x-5}-\frac{3}{3x+7}\right)dx$.

2.31 — Question 23. Show that $\int_1^\infty\frac{2}{x(3x+1)}\,dx=\ln k$, where k is a rational number to be found.

2.31 — Question 24. Showing the limiting process used, find $\displaystyle\int_{-1}^{2} \frac{2-x}{\sqrt{x+1}}\,dx$.

2.31 — Question 25. Showing the limiting process used, find $\displaystyle\int_{1/\pi}^{\infty} \frac{\sin(1/x)}{x^2}\,dx$.

2.31 — Question 26. Using the substitution $u = 2 - x^3e^{-2x}$ and showing the limiting process used, show that

$$\int_{3/2}^{\infty} \frac{(3-2x)x^2}{x^3 - 2e^{2x}}\,dx = \ln\left(\frac{16e^3}{16e^3 - 27}\right).$$

The following questions require A-Level Further Maths Integration methods, including Inverse Trigonometric Functions, Hyperbolic Functions and Partial Fractions.

2.31 — Question 27. By showing the limiting process used, show that $\displaystyle\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\,dx = \pi$.

2.31 — Question 28. Showing the limiting process used, find $\int_0^\infty (\cosh x - \sinh x)\, dx$.

2.31 — Question 29. Showing the limiting process used, find:

a) $\int_2^\infty \frac{1}{4x^2 - 9}\, dx$

b) $\int_0^\infty \frac{1}{4x^2 + 9}\, dx$

2.31 — Question 30. By showing the limiting process used, show that

$$\int_0^\infty \frac{12-6x}{(2x+3)(x^2+3)}\,dx = \ln k,$$

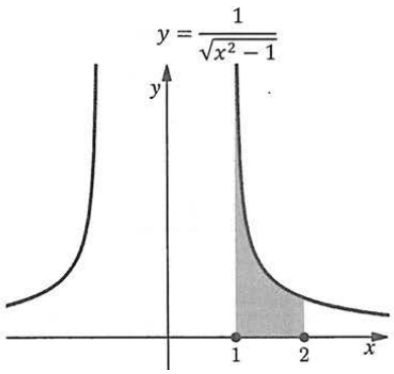
where k is a rational number to be found.

2.31 — Question 31. By showing the limiting process used, show that

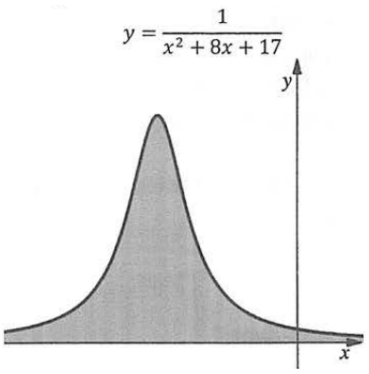
$$\int_2^\infty \frac{3x^2+16}{x^2(x^2+4)}\,dx = 2-k\pi,$$

where k is a rational number to be found.

2.31 — Question 32. The diagram shown below shows the curve $y = \dfrac{1}{\sqrt{x^2-1}}$. Find the exact shaded area between the curve, the x-axis, the line $x = 1$ and the line $x = 2$.



2.31 — Question 33. The diagram shown below shows the curve $y = \dfrac{1}{x^2+8x+17}$. Find the exact total shaded area between the curve and the x-axis.



Using L'Hôpital's Rule

2.31 — Question 34. Show that $\displaystyle\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx = -4$.

2.31 — Question 35. Showing the limiting process used, find $\displaystyle\int_0^{12} \ln x\,dx$.

2.31 — Question 36. Showing the limiting process used, find $\int_0^\infty x^2 e^{-3x}\,dx$.

2.31 — Question 37.

a) Use L'Hôpital's Rule to find $\lim_{x\to\infty}\left(\frac{\ln x}{x}\right)$.

b) Showing the limiting process used, find $\int_1^\infty \frac{\ln x}{x^2}\,dx$.

2.31 — Question 38.

a) Use L'Hôpital's Rule to find $\lim_{x\to\infty}\left(\frac{x}{e^x}\right)$.

b) Showing the limiting process used, find $\int_{-\infty}^{0} xe^x\,dx$.

2.31 — Question 39. Show clearly that $\int_{3}^{\infty} \frac{\ln x - 1}{(\ln x)^2}\,dx$ diverges.

2.32 — Question 1. Find:

a) $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{1}{1+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{x^2-1}}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{1+x^2}}\,dx$

e) $\displaystyle\int \frac{1}{1-x^2}\,dx$

2.32 — Question 2.

a) Show that $\displaystyle\int_{\pi/4}^{\pi/2} 6\cot x\,dx = \ln k$, where k is a positive integer.

b) Show that $\displaystyle\int_{0}^{\pi/2} 6\cot x\,dx$ is undefined.

2.32 — Question 3.

a) Show that the substitution $u = \tan x$ transforms $\displaystyle\int \frac{1}{25\cos^2 x + \sin^2 x}\,dx$ into $\displaystyle\int \frac{1}{25 + u^2}\,du$.

b) Hence show that $\displaystyle\int_0^{\arctan 5} \frac{1}{25\cos^2 x + \sin^2 x}\,dx = k\pi$, where k is a rational number to be found.

2.32 — Question 4. Show that $\displaystyle\int_{\sqrt{\ln 2}}^{\sqrt{\ln 5}} \frac{x}{\sinh(x^2)}\,dx = \frac{1}{2}\ln 2$.

2.32 — Question 5.

a) Find $\displaystyle\int \frac{1}{x^2 - 12x + 61}\, dx$.

b) Hence find the exact value of $\displaystyle\int_5^7 \left(3 - \frac{78}{x^2 - 12x + 61}\right) dx$.

c) The curve $y = 3 - \dfrac{78}{x^2 - 12x + 61}$ intersects the x-axis at the points P and Q.

a) Find the coordinates of P and Q.

b) The curve has a stationary point at the point R. Find the coordinates of R.

c) Find the exact area of the region between the curve, the x-axis, the line $x = 5$ and the line $x = 6$.

2.32 — Question 6.

a) Find $\displaystyle\int \frac{2x^3 + x^2 + x}{(x+1)(x^2+1)}\, dx$.

b) Hence show that $\displaystyle\int_0^1 \frac{2x^3 + x^2 + x}{(x+1)(x^2+1)}\, dx = 2 - \ln 2 - \frac{\pi}{4}$.

2.32 — Question 7. By using a suitable substitution, show that:

a) $\displaystyle\int_{-5/2}^{-3/2} \frac{1}{\sqrt{4x^2+20x+34}}\,dx = \frac{1}{2}\ln\left(\frac{2+\sqrt{13}}{3}\right)$

b) $\displaystyle\int_{1/2}^{5/4} \frac{1}{\sqrt{-4x^2+4x+8}}\,dx = \frac{\pi}{12}$

c) $\displaystyle\int_{(\sqrt{3}-3)/2}^{0} \frac{1}{\sqrt{4x^2+12x+18}}\,dx = \frac{\pi}{72}$

d) $\displaystyle\int_{3}^{4} \frac{1}{\sqrt{-4x^2+28x-40}}\,dx = \frac{1}{6}\ln 2$

2.32 — Question 8. Show that $\displaystyle\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{12x^2}{1-x^4}\,dx = 3\ln(p+\sqrt{q}) + r\pi$, where p, q and r are integers to be found.

2.33 — Question 1. If $I_n = \int x^n e^x \, dx$, show that for $n \geq 1$,

$$I_n = x^n e^x - nI_{n-1}.$$

2.33 — Question 2. For $I_n = \int x^n e^x \, dx$ it is given that $I_n = x^n e^x - nI_{n-1}$ for $n \geq 1$. Evaluate each of the following:

a) $\int x^2 e^x \, dx$

b) $\int x^3 e^x \, dx$

c) $\int x^4 e^x \, dx$

2.33 — Question 3. If $I_n = \int x^n e^{2x}\,dx$, show that for $n \geq 1$,

$$I_n = \frac{1}{2}x^n e^{2x} - \frac{n}{2}I_{n-1}.$$

2.33 — Question 4. For $I_n = \int x^n e^{2x}\,dx$ it is given that $I_n = \frac{1}{2}x^n e^{2x} - \frac{n}{2}I_{n-1}$ for $n \geq 1$. Evaluate each of the following:

a) $\int x^3 e^{2x}\,dx$

b) $\int x^4 e^{2x}\,dx$

2.33 — Question 5. If $I_n = \int x(\ln x)^n \, dx$, show that for $n \geq 2$,

$$I_n = \frac{x^2}{2}(\ln x)^n - \frac{n}{2}I_{n-1}.$$

2.33 — Question 6. For $I_n = \int x(\ln x)^n \, dx$ it is given that $I_n = \frac{x^2}{2}(\ln x)^n - \frac{n}{2}I_{n-1}$ for $n \geq 2$. Evaluate each of the following:

a) $\int x(\ln x)^2 \, dx$

b) $\int x(\ln x)^3 \, dx$

2.33 — Question 7. If $I_n = \int \sin^n x \, dx$, show that for $n \geq 2$,

$$I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}.$$

2.33 — Question 8. For $I_n = \int \sin^n x \, dx$ it is given that $I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$ for $n \geq 2$. Evaluate each of the following:

a) $\int \sin^4 x \, dx$

b) $\int \sin^5 x \, dx$

2.33 — Question 9. If $I_n = \int \cos^n x \, dx$, show that for $n \geq 2$,

$$I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}.$$

2.33 — Question 10. For $I_n = \int \cos^n x \, dx$ it is given that $I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$ for $n \geq 2$. Evaluate each of the following:

a) $\int \cos^4 x \, dx$

b) $\int \cos^5 x \, dx$

Prerequisite Knowledge for Questions 11–12: Calculus of Hyperbolic Functions

2.33 — Question 11. Let $I_n = \int \sinh^n x \, dx$.

a) Show that for $n \geq 2$, $I_n = \dfrac{\sinh^{n-1} x \cosh x}{n} - \dfrac{n-1}{n} I_{n-2}$.

b) Hence evaluate $\int \sinh^4 x \, dx$.

2.33 — Question 12. Let $I_n = \int \cosh^n x \, dx$.

a) Show that for $n \geq 2$, $I_n = \dfrac{\cosh^{n-1} x \sinh x}{n} + \dfrac{n-1}{n} I_{n-2}$.

b) Hence evaluate $\int \cosh^6 x \, dx$.

2.33 — Question 13. Let $I_n = \int x^n \sqrt{1+x}\, dx$.

a) Show that for $n \geq 1$,

$$I_n = \frac{2x^n(1+x)^{3/2}}{2n+3} - \frac{2n}{2n+3} I_{n-1}.$$

b) Hence evaluate $\int x^2 \sqrt{1+x}\, dx$.

2.33 — Question 14. Let $I_n = \int \tan^n x\, dx$.

a) By considering $1 + \tan^2 x = \sec^2 x$, show that for $n \geq 2$,

$$I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}.$$

b) Show that $\int \tan x\, dx = \ln|\sec x| + c$.

c) Hence evaluate $\int \tan^5 x\, dx$.

2.33 — Question 15. Let $I_{m,n} = \int x^m (\ln x)^n \, dx$.

a) Show that for $n \geq 1$,

$$I_{m,n} = \frac{x^{m+1}(\ln x)^n}{m+1} - \frac{n}{m+1} I_{m,n-1}.$$

b) Hence evaluate $\int x^3 \ln x \, dx$.

Definite Integrals

2.33 — Question 16. Let $I_n = \int_0^1 \frac{x^n}{\sqrt{1-x}} \, dx$.

a) Show that for $n \geq 1$, $I_n = \frac{2n}{1+2n} I_{n-1}$.

b) Hence show that $\int_0^1 \frac{x^4}{\sqrt{1-x}} \, dx = \frac{256}{315}$.

2.33 — Question 17. Let $I_n = \int_0^{\pi/2} x^n \sin x \, dx$.

a) Show that for $n \geq 2$, $I_n = n\left(\frac{\pi}{2}\right)^{n-1} - n(n-1)I_{n-2}$.

b) Hence show that $\int_0^{\pi/2} x^4 \sin x \, dx = \frac{1}{2}(48 - 24\pi + \pi^3)$.

2.33 — Question 18. Let $I_n = \int_0^{2\pi} \cos^n x \, dx$. By considering the reduction formula in each case, show that when:

a) n is odd, $I_n = 0$

b) n is even, $I_n = \left(\frac{2n-1}{2n}\right)\left(\frac{2n-3}{2n-2}\right)\cdots\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)2\pi$.

2.33 — Question 19. Let $I_n = \int_0^{2\pi} \sin^n x \, dx$. By considering the reduction formula in each case, show that when:

a) n is odd, $I_n = 0$

b) n is even, $I_n = \left(\frac{2n-1}{2n}\right)\left(\frac{2n-3}{2n-2}\right)\cdots\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)2\pi$.

2.33 — Question 20. Let $I_n = \int_0^{\pi/3} \sec^n x \, dx$.

a) Show that for $n \geq 1$, $(n-1)I_n = 2^{n-2}\sqrt{3} + (n-2)I_{n-2}$.

b) Hence evaluate $\int_0^{\pi/3} \sec^8 x \, dx$.

2.33 — Question 21. Let $I_n = \int_0^{\pi/3} \sin^n x \, dx$.

a) Show that for $n \geq 2$, $I_n = \frac{n-1}{n} I_{n-2} - \frac{1}{2n}\left(\frac{\sqrt{3}}{2}\right)^{n-1}$.

b) Hence show that $\int_0^{\pi/3} \sin^6 x \, dx = a\pi - b\sqrt{3}$, where a and b are constants to be found.

c) State the value of $\int_{-\pi/3}^{\pi/3} \sin^6 x \, dx$.

2.33 — Question 22. Let $I_n = \int_0^{\pi/2} \cos^n x \, dx$. Given that for $n \geq 2$, $I_n = \frac{n-1}{n} I_{n-2}$, evaluate $\int_{-\pi/2}^{\pi/2} \cos^6(3t) \, dt$.

2.33 — Question 23. Let $I_n = \int_0^1 (1-x^2)^n \, dx$.

a) Show that for $n \geq 1$, $I_n = \dfrac{2n}{2n+1} I_{n-1}$.

b) Hence evaluate $\int_{-1}^1 (1-x^2)^6 \, dx$.

2.33 — Question 24. Given that $n \geq 2$, if $I_n = \int_0^{\pi/2} \cos^n x \, dx$ then $I_n = \dfrac{n-1}{n} I_{n-2}$.

a) Show that for $n \geq 1$, $I_n = \dfrac{2n-1}{2n+2} I_{n-1}$.

b) Hence show that $\int_0^{\pi/2} \cos^{12} x \sin^2 x \, dx = \dfrac{33\pi}{4096}$.

c) State the value of $\int_{-\pi/2}^{\pi/2} \cos^{12} x \sin^2 x \, dx$.

Prerequisite Knowledge for Question 25: Calculus of Hyperbolic Functions

2.33 — Question 25. Let $I_n = \displaystyle\int_0^{\ln 3} \tanh^n x \, dx$.

a) By considering $\tanh^2 x + \operatorname{sech}^2 x = 1$, show that for $n \geq 2$,

$$I_n = I_{n-2} - \frac{1}{n-1}\left(\frac{4}{5}\right)^{n-1}.$$

b) Hence show that $\displaystyle\int_0^{\ln 3} \tanh^4 x \, dx = \ln a - \frac{b}{c}$, where a, b and c are integers to be found.

2.33 — Question 26. Let $I_n = \displaystyle\int_0^1 x^n e^{ax} \, dx$ for constant $a \neq 0$.

a) Show that for $n \geq 1$, $I_n = \dfrac{e^a}{a} - \dfrac{n}{a} I_{n-1}$.

b) Hence determine the value of $\displaystyle\int_0^1 x^2 e^{x/4} \, dx$.

2.33 — Question 27. Let $I_n = \int_0^{\sqrt{a}} (a - x^2)^n \, dx$ where a is a positive constant.

a) Show that for $n \geq 1$, $I_n = \dfrac{2na}{2n+1} I_{n-1}$.

b) Hence evaluate $\int_0^4 (16 - x^2)^3 \, dx$.

2.33 — Question 28. Let $I_n = \int_0^{\pi} \dfrac{\cos nx}{\cos x} \, dx$.

a) Given $\cos(A + B) + \cos(A - B) = 2\cos A \cos B$, show that

$$\frac{\cos nx}{\cos x} = 2\cos((n-1)x) - \frac{\cos((n-2)x)}{\cos x}.$$

b) Show that for $n \geq 2$, $I_n = -I_{n-2}$.

c) Hence show that $\int_0^{\pi} \dfrac{\cos 7x}{\cos x} \, dx = -\pi$.

d) Evaluate $\int_0^{\pi} \dfrac{\cos 53x}{\cos x} \, dx$.

2.33 — Question 29. Let $I_{m,n} = \int_0^k x^n(a-x)^m\,dx$ for constant $k < a$.

a) Show that for $n \geq 1$,

$$I_{m,n} = -\frac{k^n(a-k)^{m+1}}{m+1} + \frac{n}{m+1}I_{m+1,n-1}.$$

b) Hence show that $\int_0^1 \frac{x^4}{(2-x)^3}\,dx = a\ln 2 - b$, where a and b are positive constants to be found.

Prerequisite Knowledge for Question 30: L'Hôpital's Rule and Improper Integrals

2.33 — Question 30. Let $I_n = \int_0^1 (\ln x)^n\,dx$.

a) Find $\lim_{x\to 0^+}(x\ln x)$.

b) Show that for $n \geq 1$, $I_n = -nI_{n-1}$.

c) Hence show that for $n \geq 1$, $I_n = (-1)^n n!$.

d) Evaluate $\int_0^1 (\ln x)^4\,dx$.